QPCODE: 25900319 – Scheme of Valuation

MGU BCA October 2025

Third Semester DSE- MG3DSEBCA201- Feature Engineering

Part A (Short Answer) Any 10 questions 10X2 =20 Marks

1. Why can't mathematical operations be directly applied to categorical data?

Categorical data represents labels or categories, not numerical quantities. They do not have inherent numeric value or order (no meaningful arithmetic relationship). Operations like addition or subtraction do not provide meaningful results for categories. Categorical variables are qualitative, so statistical/mathematical operations cannot interpret category differences correctly. Any 2 points- **2 marks.**

2. Define continuous data and give finance related example.

Definition (**1 Mark**): Continuous data refers to numerical data that can take any value within a given range, including decimals and fractions.

Finance-related Example (**1 Mark**): Examples: Stock prices, interest rates, currency exchange rates, revenue values, share market index values, etc.

3. Define Data with an example

Definition (**1 Mark**): Data refers to raw facts, figures, or observations that can be collected, stored, and processed to extract meaningful information.

Example (**1 Mark**): Examples: List of student ages, daily temperatures, product sales numbers, survey responses, etc.

4. Why data cleaning important in feature engineering?

Removes noise, errors, and inconsistencies from data to improve model performance. Ensures high-quality and reliable features for accurate predictions. Helps handle missing values and outliers that could mislead the model. Improves training efficiency and reduces bias in machine learning models. Any two points- **2 Marks.**

5. Why does normalization improve the performance of distance related algorithms?

Normalization scales all features to a similar range, preventing features with larger values from dominating distance calculations. It ensures fair contribution of all features in algorithms like KNN and K-Means. Helps the model converge faster and improves accuracy and stability of results. Reduces the bias caused by varying units and magnitudes of features. Any two points -**2 Marks**

6. What is handling missing data?

Handling missing data refers to the process of identifying and treating absent or incomplete values in a dataset. It ensures data quality and prevents skewed or incorrect results in analysis or machine learning. Techniques include removal of missing entries, imputation, or using model-based methods to fill gaps. Any two points- **2 Marks.**

7. Write the formula to calculate bin width in equal width discretization. **2 Marks**

$$\text{Bin Width} = \frac{\text{Max Value} - \text{Min Value}}{\text{Number of Bins}}$$

8. What is the purpose of using polynomial and interaction features in machine learning models?

They allow models to capture non-linear relationships between features and the target. Improve model accuracy by expanding the feature space. Interaction features help identify combined effects of multiple variables. Enhance predictive power without changing the algorithm. Any two points -**2 Marks.**

9. List the main steps in feature selection.

Identify relevant features from the dataset. Apply feature selection techniques (filter, wrapper, or embedded methods). Evaluate model performance with selected features. Remove irrelevant, redundant, or highly correlated features. Any two points- **2 Marks.**

10. Compare the practical advantages and limitations of filter wrapper and hybrid feature selection methods with suitable examples.

Filter methods use statistical techniques to select features before applying any model. Advantage: Fast and useful for high-dimensional datasets. . Limitation: Selection may not always improve performance of a particular model since they ignore model behaviour. Example: Correlation-based feature selection, Chi-square test. Wrapper Methods evaluate subsets of features by training and testing a model repeatedly. Advantage: Produces features specific to the model, improving accuracy. Limitation: Very slow and computationally expensive; may leads to overfitting. Example: Recursive Feature Elimination (RFE**)** with SVM or Logistic Regression. Hybrid Methods combine filter and wrapper methods to balance speed and accuracy. Advantage: More efficient than wrapper and more accurate than filter alone. Limitation: More complex to design and implement. Example: Using a filter method to reduce dimensionality first, then applying RFE. Any two points – **2 Marks.**

Part B. Answer any 3 questions. 3 X 5 = 15 Marks

11. Explain with example how discrete data is different from continuous data.

Definition of Discrete Data: Data that consists of distinct, separate values; countable; no intermediate values possible **1 Mark**.

Example of Discrete Data: Number of students in a class, number of cars, etc. **1 Mark**.

Definition of Continuous Data: Data that can take any value within a range; measurable; includes decimals/fractions **1 Mark**.

Example of Continuous Data- Height, weight, temperature, stock prices, etc.**1 Mark**.

Any one relevant comparison - Discrete → countable; Continuous → measurable, Discrete values cannot be subdivided meaningfully; Continuous values can, Discrete shown as bar graphs; continuous shown as histograms **1 Mark.**

12. Define feature scaling. List and explain different methods of feature scaling with suitable formulas and examples.

Definition of Feature Scaling: Process of transforming features into a similar numerical scale to improve model performance (especially in distance-based ML models). **1 Mark**.

Any Two methods explained with formula and example- **2 Marks each**.

13. Discuss the importance of interaction features in real world applications. Provide two examples.

Definition / Meaning of Interaction Features: Interaction features capture combined effects of two or more features that influence the target variable together. **1 Mark**.

Importance in Real-World Applications (Any three valid points**- 3 Marks**). Helps discover non-linear relationships between variables. Improves model accuracy and predictive power. Reveals hidden patterns not captured by individual features. Essential in domains like marketing, finance, medicine where combined influence matters. Reduces model bias by including real-world variable interactions.

Any two examples such as: Marketing: Interaction of Income × Age to predict buying behaviour, Healthcare: BP × Cholesterol interaction for heart disease risk, E-commerce: Discount × User Loyalty affecting purchase likelihood, Real Estate: Area × Location Rating predicting house prices. **1 Mark**.

14. Compare the advantages and disadvantages of label encoding with examples.

Definition of Label Encoding. Converting categorical values into numeric labels (e.g., A → 0, B → 1). **2 Mark**.

Advantages (Any Two- **1 Mark**) Simple and quick to implement. Useful for ordinal data where order is important. Efficient storage and model processing due to numeric form.

Disadvantages (Any Two- **1 Mark**) Implies false ordinal relationships between categories. Can mislead distance-based and tree-based models. Different numeric labels may affect model output unintentionally.

Example with categories- **1 Mark**: Example: Colors: Red, Blue, Green → 0, 1, 2 or Education Level: High School → 0, UG → 1, PG → 2 (valid ordinal case)

15. Explain with reasoning why PCA may reduce accuracy in classification tasks.

PCA optimizes variance, not class separation-PCA finds orthogonal directions that capture the most data variance. If class-distinguishing information lies in directions with low overall variance, PCA may discard that information when you reduce dimensionality.

Discriminative signal may be low-variance.

Linear projection assumption. Standard PCA is linear. If classes are separable only by nonlinear structure (manifold), PCA's linear subspace projection will mix and collapse those nonlinear distinctions.

Mixing of features reduces interpretability. Principal components are linear combinations of original features. If a few original features were directly predictive, PCA spreads that information across components; truncation can therefore lose the predictive contribution.

Scale and preprocessing sensitivity. If features aren't properly scaled, PCA can emphasize large-magnitude features regardless of their predictive power, harming classifiers that expect balanced inputs.

Class imbalance and noise issues. Noise or a dominant majority class can drive variance directions. PCA might then prioritize noise or majority-class variation, degrading performance on minority classes. **Any five valid points – 5 Marks**